AI-Autostart User Manual

Abstract of the system:

AI-Autostart is a gesture-based desktop automation system designed to enable hands-free interaction with applications, games, and files through real-time hand gesture recognition. The system runs in the background using a system tray interface, allowing users to trigger actions such as opening applications, launching games and executing files.

The application provides a configurable settings interface where users can assign gestures to specific actions, manage application and game mappings, enable or disable camera preview, and adjust recognition features such as person detection. AI-Autostart is designed to be lightweight, responsive, and user-friendly.

This system aims to improve accessibility, enhance user interaction with computing environments, and provide an intuitive alternative to traditional input methods such as keyboards and mice.

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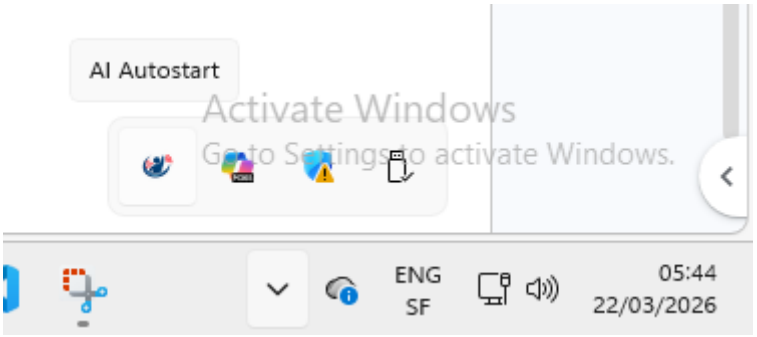
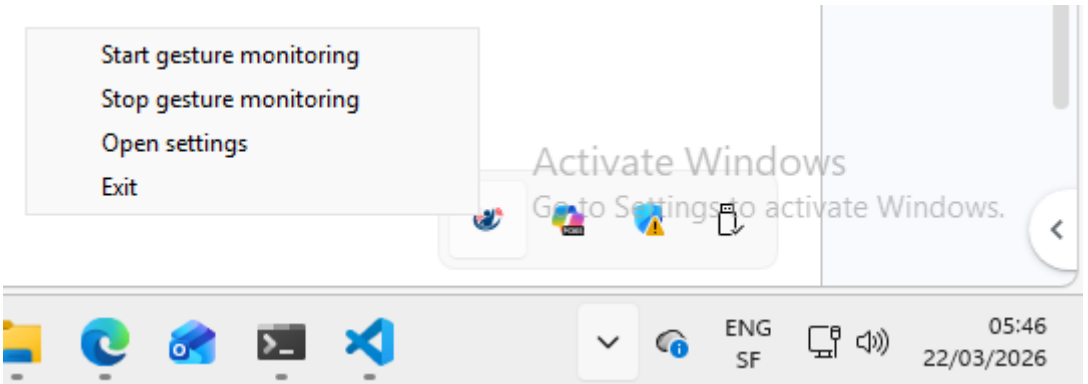
1. Getting Started

AI-Autostart is a desktop application that runs in the system tray and allows users to control applications using hand gestures detected through a webcam. Once the application is active, it appears in the bottom taskbar of the computer, usually inside the hidden icons section. Hovering over the icon displays the text **AI-Autostart**.



Right-clicking the AI-Autostart icon opens a menu that provides access to gesture monitoring, settings, and application controls.

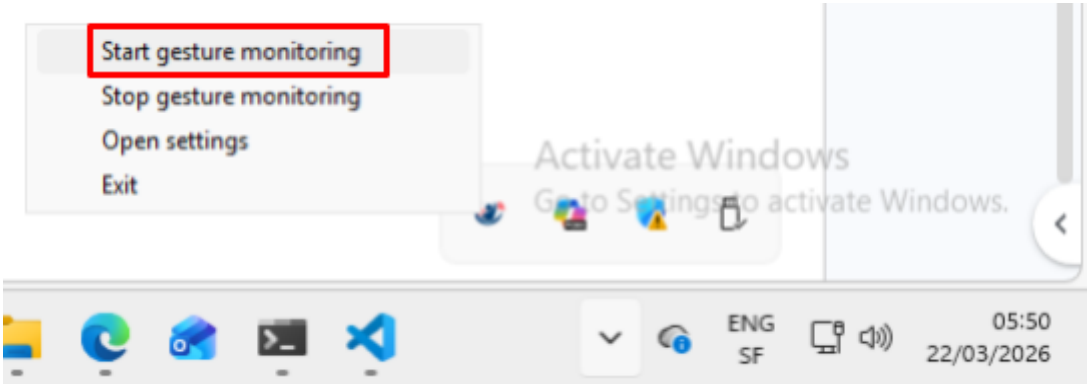


2. Starting Gesture Monitoring

To start gesture monitoring:

1. Launch AI-Autostart.
2. Locate the AI-Autostart icon in the system tray.
3. Right-click the icon.
4. Click **Start gesture monitoring**.

After clicking this option, the camera turns on and gesture monitoring begins.



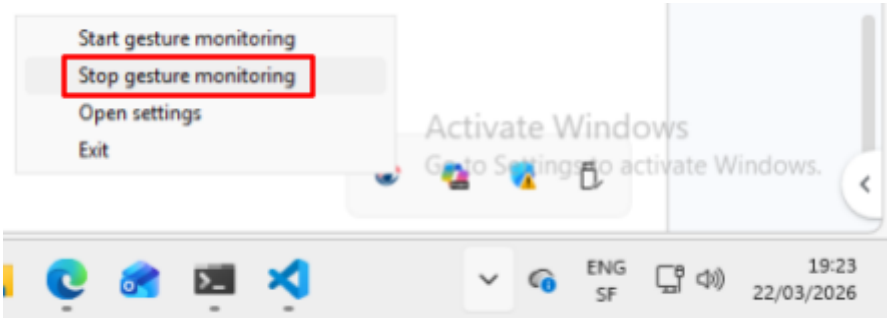
3. Stopping Gesture Monitoring

Gesture monitoring can be stopped in two ways.

Option 1: Using the System Tray

1. Right-click the AI-Autostart icon.
2. Click **Stop gesture monitoring**.

The camera turns off and gesture monitoring stops.



Option 2: Using a Gesture

1. Perform the gesture assigned to stopping the gesture recognizer.
2. The default gesture is **I Love You Left**.
3. This gesture can be changed in the Settings page.

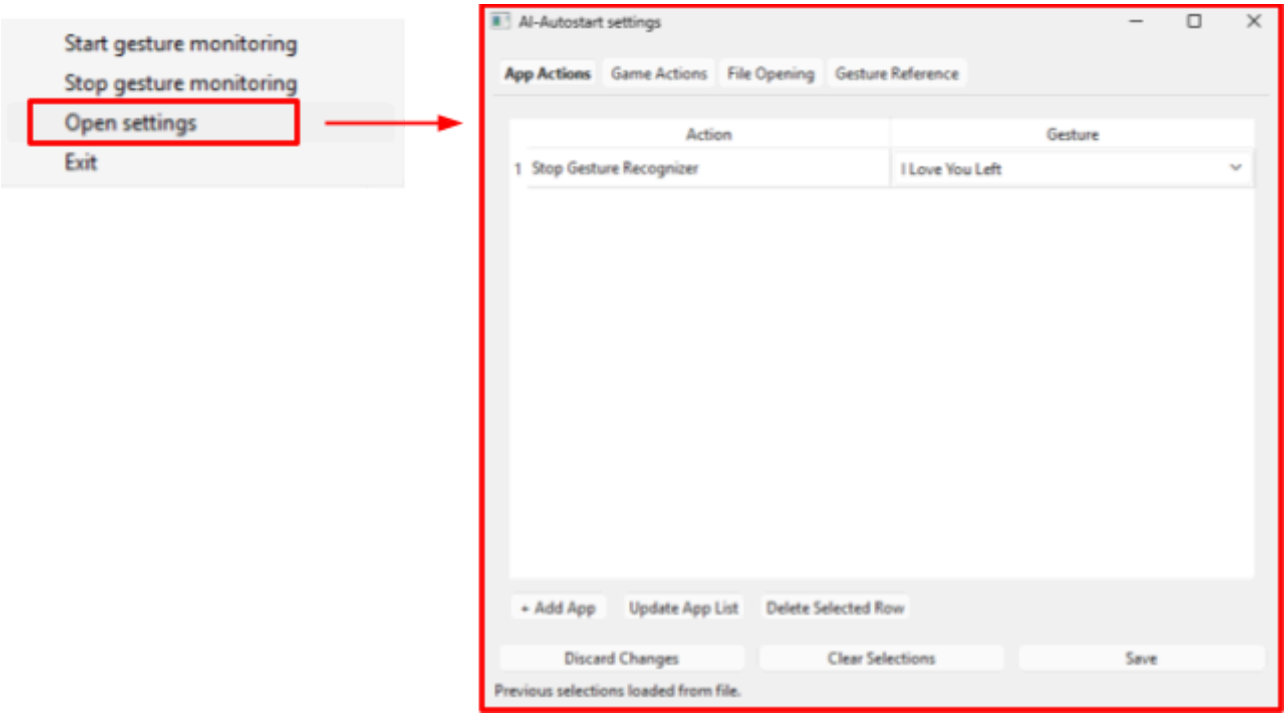
Once the gesture is performed, the camera turns off and monitoring stops.

4. Configuring AI-Autostart Settings

Opening the Settings Page

1. Right-click the AI-Autostart icon in the system tray.
2. Click **Open settings**.

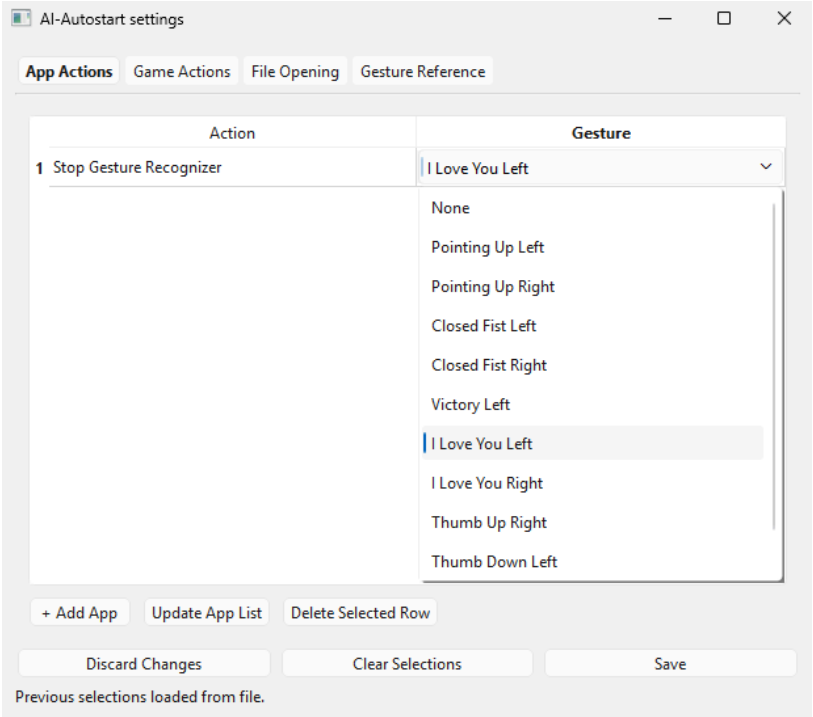
The settings window will appear, allowing configuration of gestures, applications, games, and files.



5. Changing Gesture Mappings

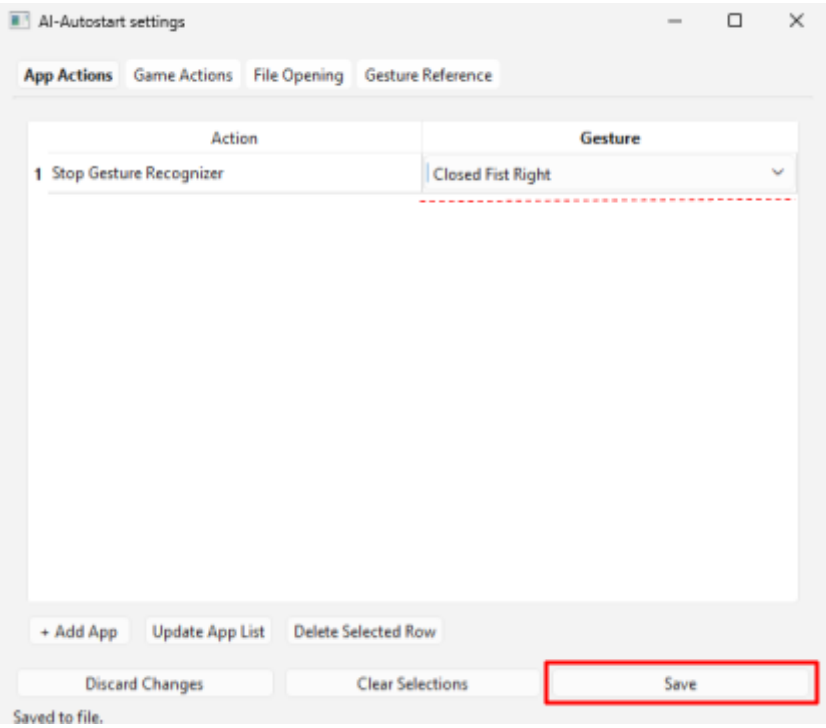
To change the gesture assigned to an action:

1. Click the current gesture name in the **Gesture** column.
2. A dropdown menu with available gestures will appear.



3. Select the desired gesture.
4. Click **Save** in the bottom-right corner of the window.

A confirmation message will appear at the bottom of the window indicating that the changes have been saved.



Important Notes

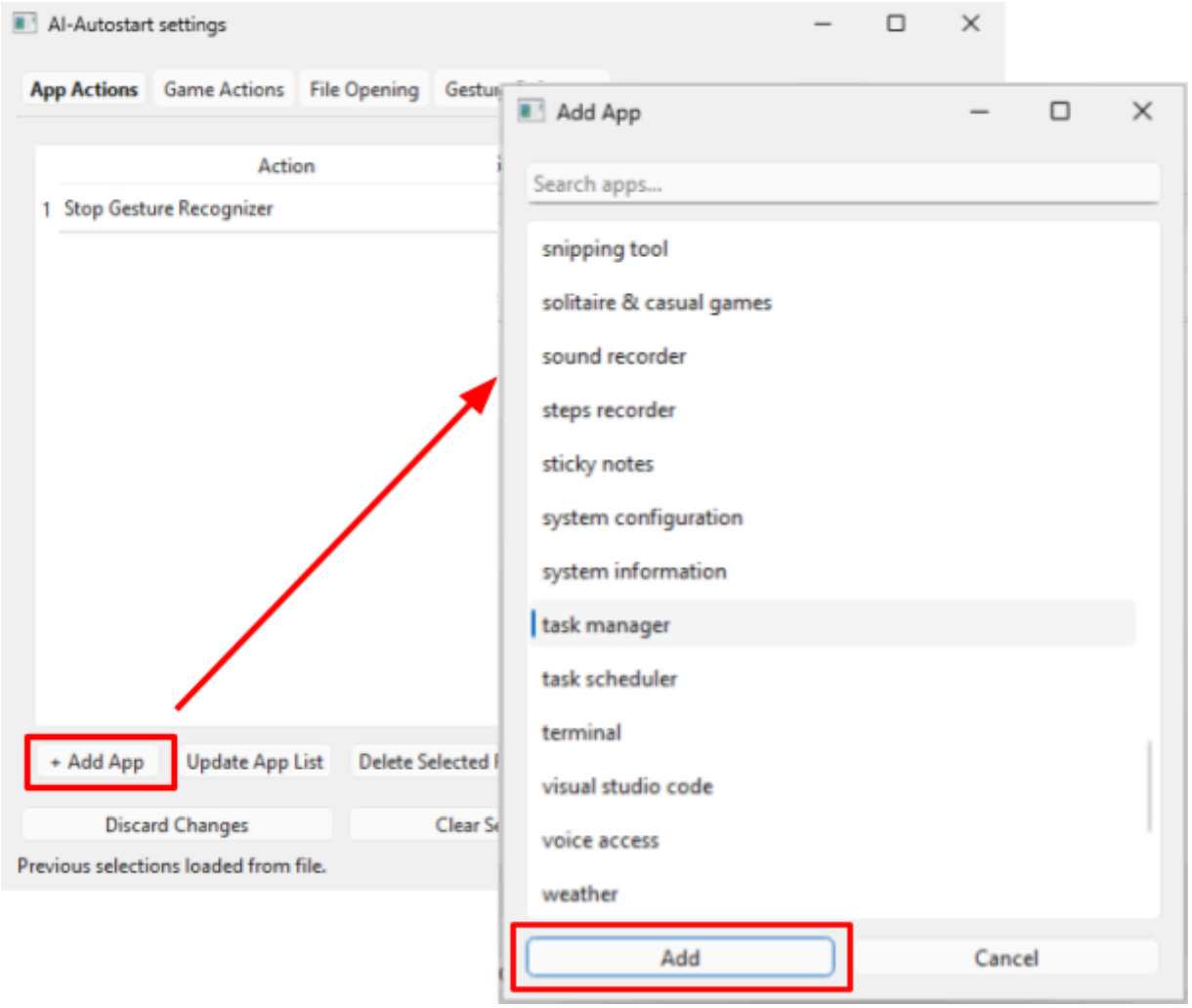
- Saved changes are applied automatically if gesture monitoring is running.
- Restarting the application is not required.
- Each gesture can only be assigned to one action at a time.
- If a gesture does not appear in the dropdown menu, it is already assigned to another action.
- Remove it from the other action to make it available.

6. Configuring App Actions

6.1 Adding an App

To configure a new application:

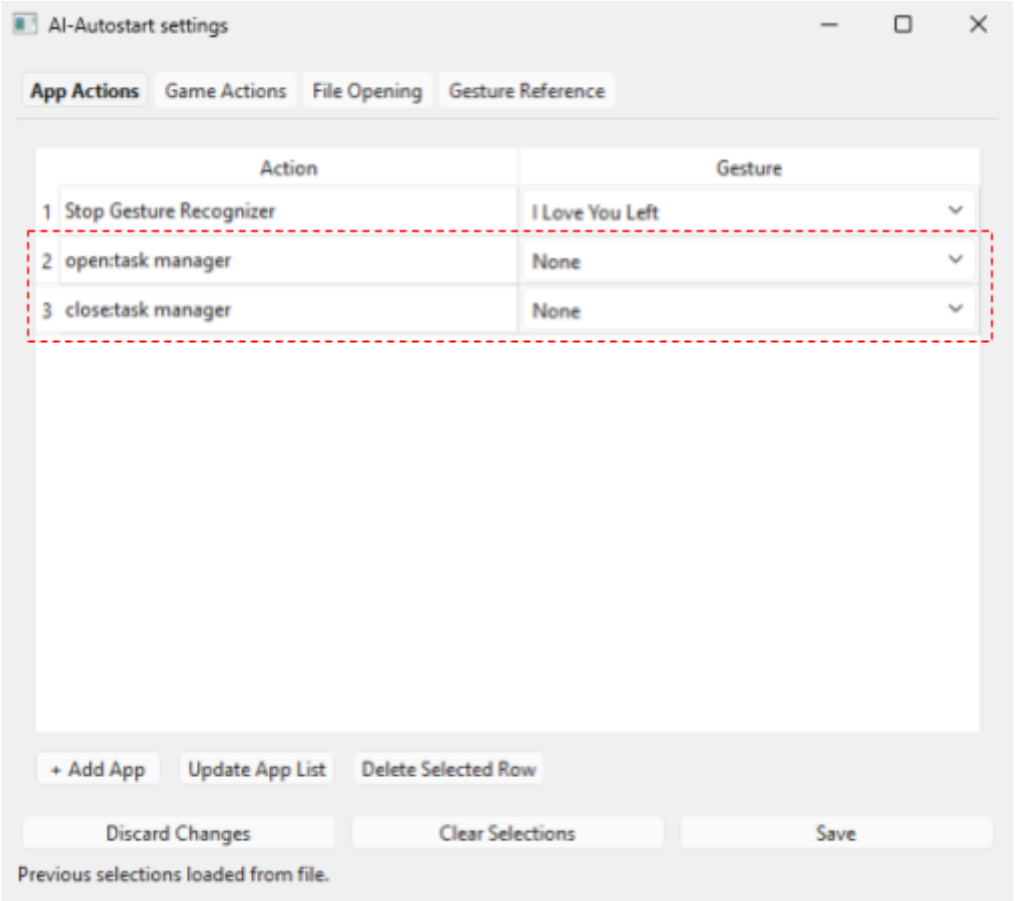
1. Click **Add App** in the bottom-left corner of the settings window.
2. A list of installed applications will appear.
3. Select the desired application.
4. Click **Add**.



Two new rows will be added to the table:

- Open application
- Close application

You can now assign gestures to these actions.


AI-Autostart settings
App Actions
Game Actions
File Opening
Gesture Reference
Action
Gesture
1 Stop Gesture Recognizer
I Love You Left
2 open:task manager
None
3 close:task manager
None
+ Add App
Update App List
Delete Selected Row
Discard Changes
Clear Selections
Save
Previous selections loaded from file.

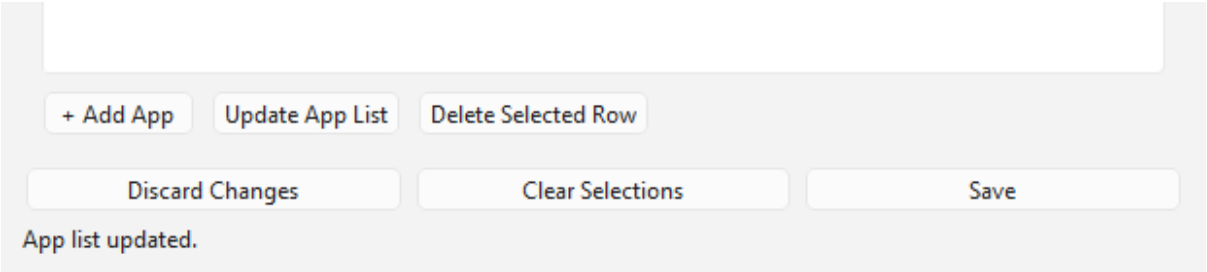
6.2 Updating the App List

The application list updates automatically each time the settings page is opened.

If a new application is installed while the settings page is open:

1. Click **Update App List** at the bottom of the window.
2. Wait for the confirmation message.

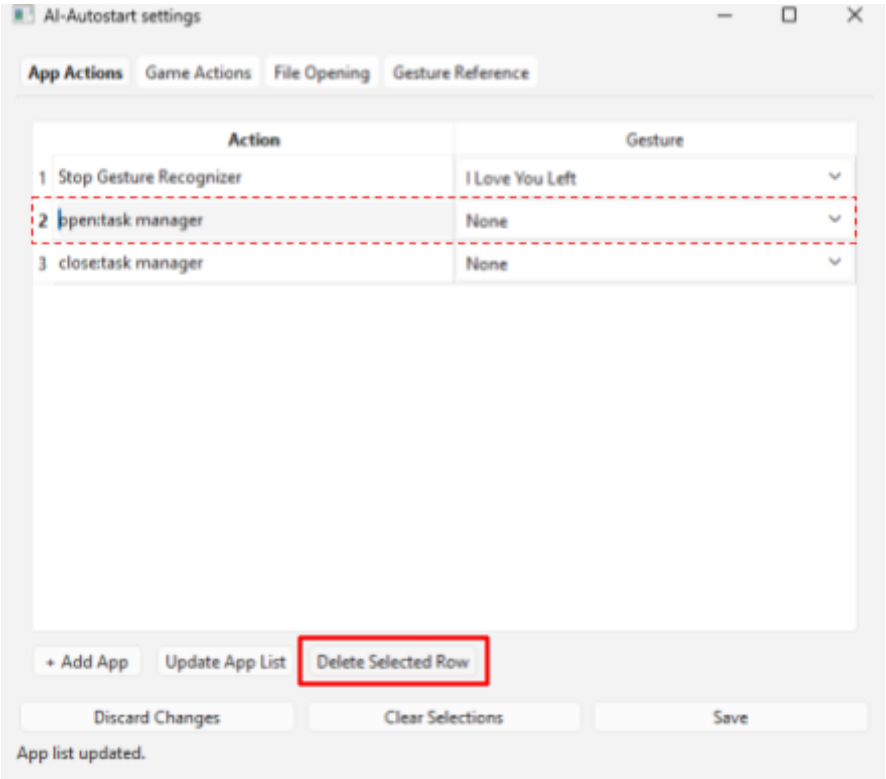
The list will refresh and include newly installed applications.



6.3 Deleting a Table Row

To delete a row:

1. Select a row by clicking on it in the **Action** column.
2. Click **Delete Selected Row** at the bottom of the window.



3. Click **Save**.

The selected row will be removed from the table.

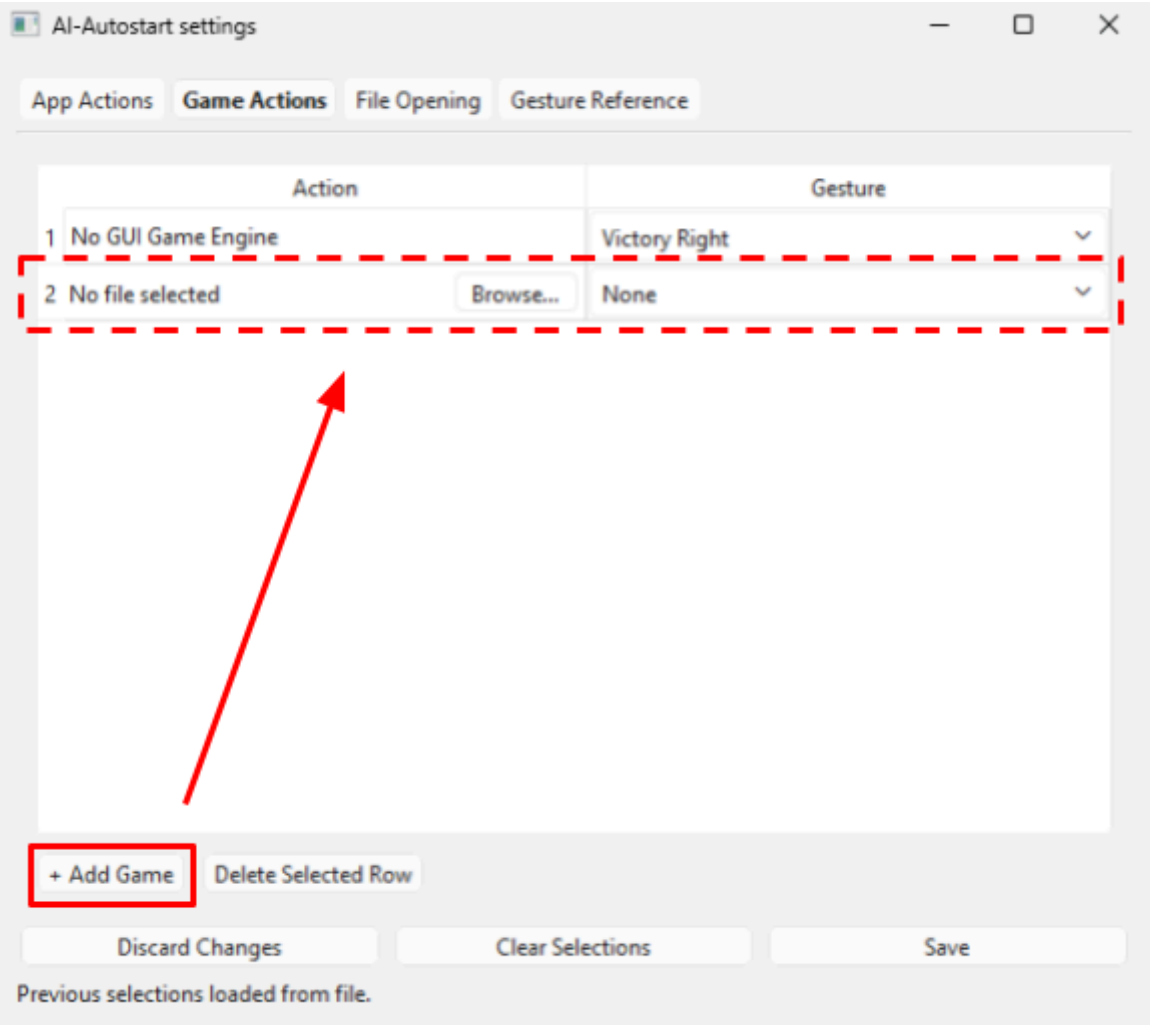
Note: The **Stop Gesture Recognizer** row cannot be deleted.

7. Configuring Game Actions

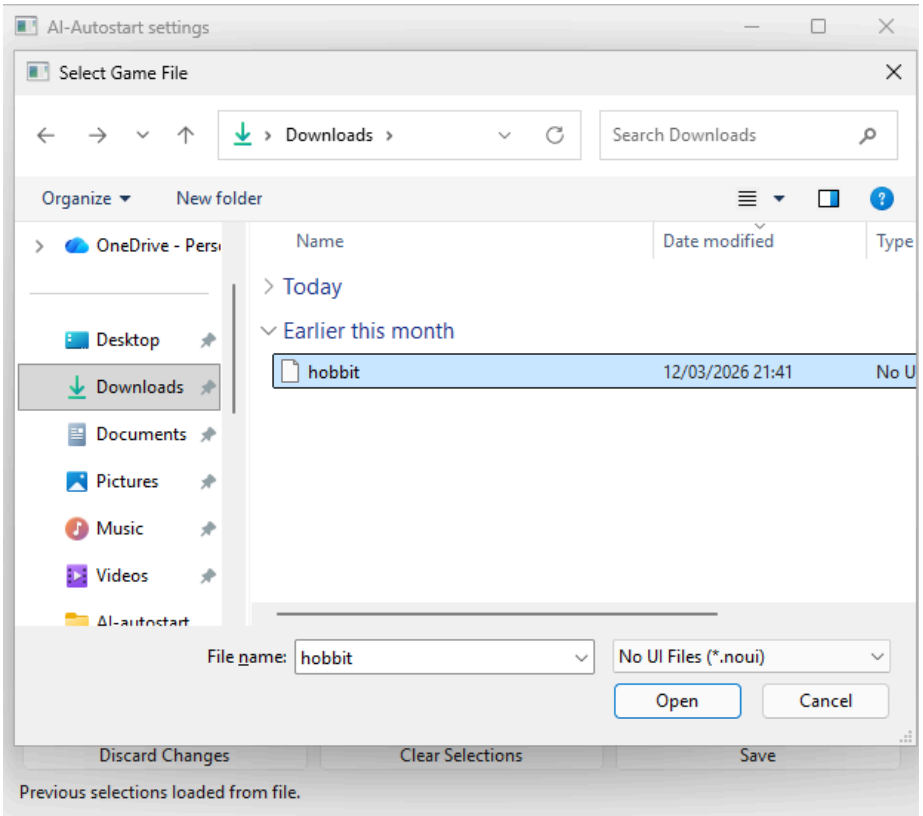
Game Actions allow users to launch games using gestures.

7.1 Adding a Game

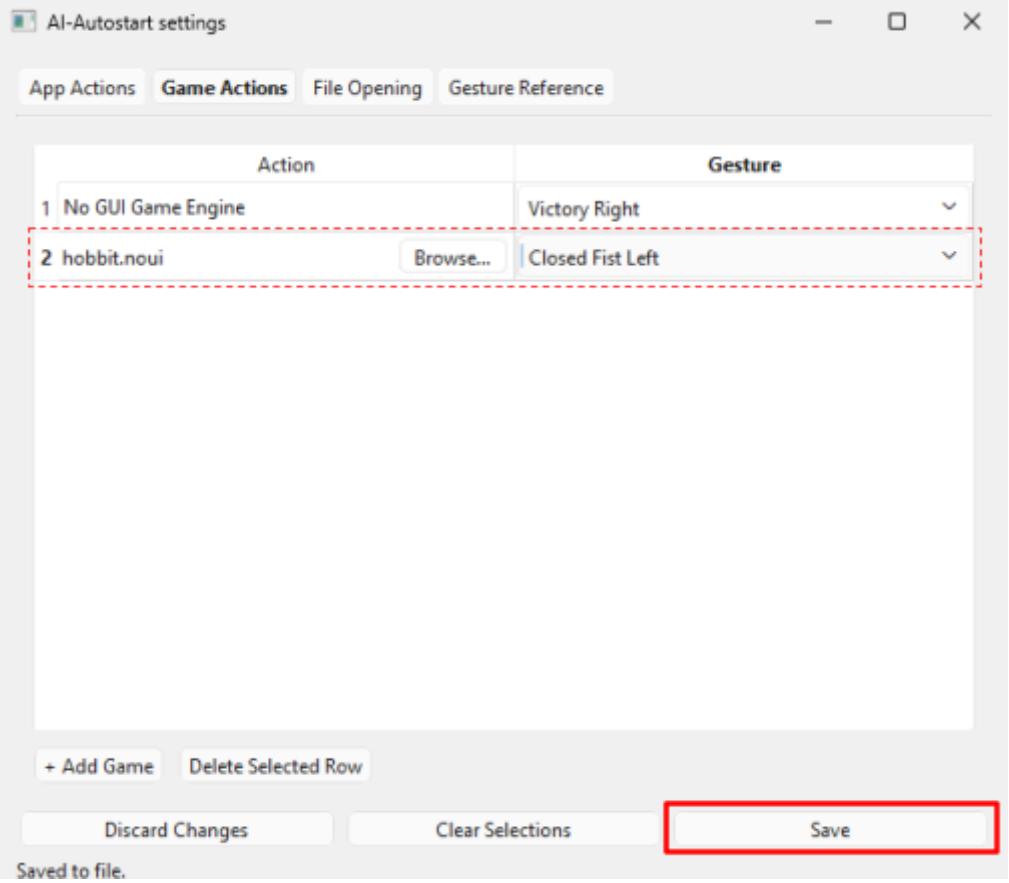
1. Open the settings window from the tray menu.
2. Go to **Game Actions**.
3. Click + **Add Game**.
4. A new row will appear.



5. Click **Browse** and select the game file (the dialog is optimized for .noui file types).



6. Choose a gesture in the **Gesture** column.
7. Click **Save**.



Note:
If the gesture is set to **None**, the game row is inactive.

7.2 Deleting a Game

1. Select the game row.
2. Click **Delete Selected Row**.
3. Click **Save**.

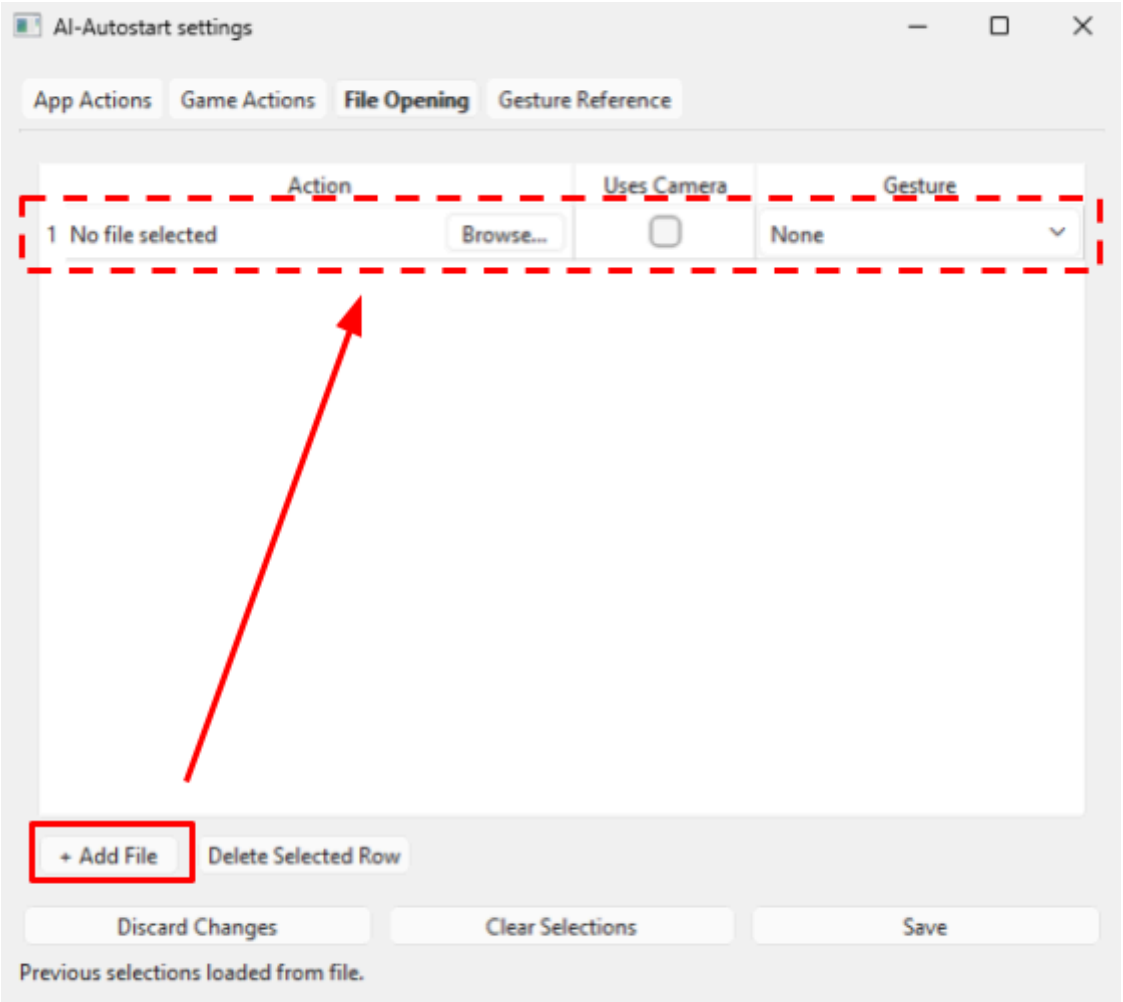
Note:
The first row (**No GUI Game Engine**) is built-in and cannot be deleted.

8. File Opening

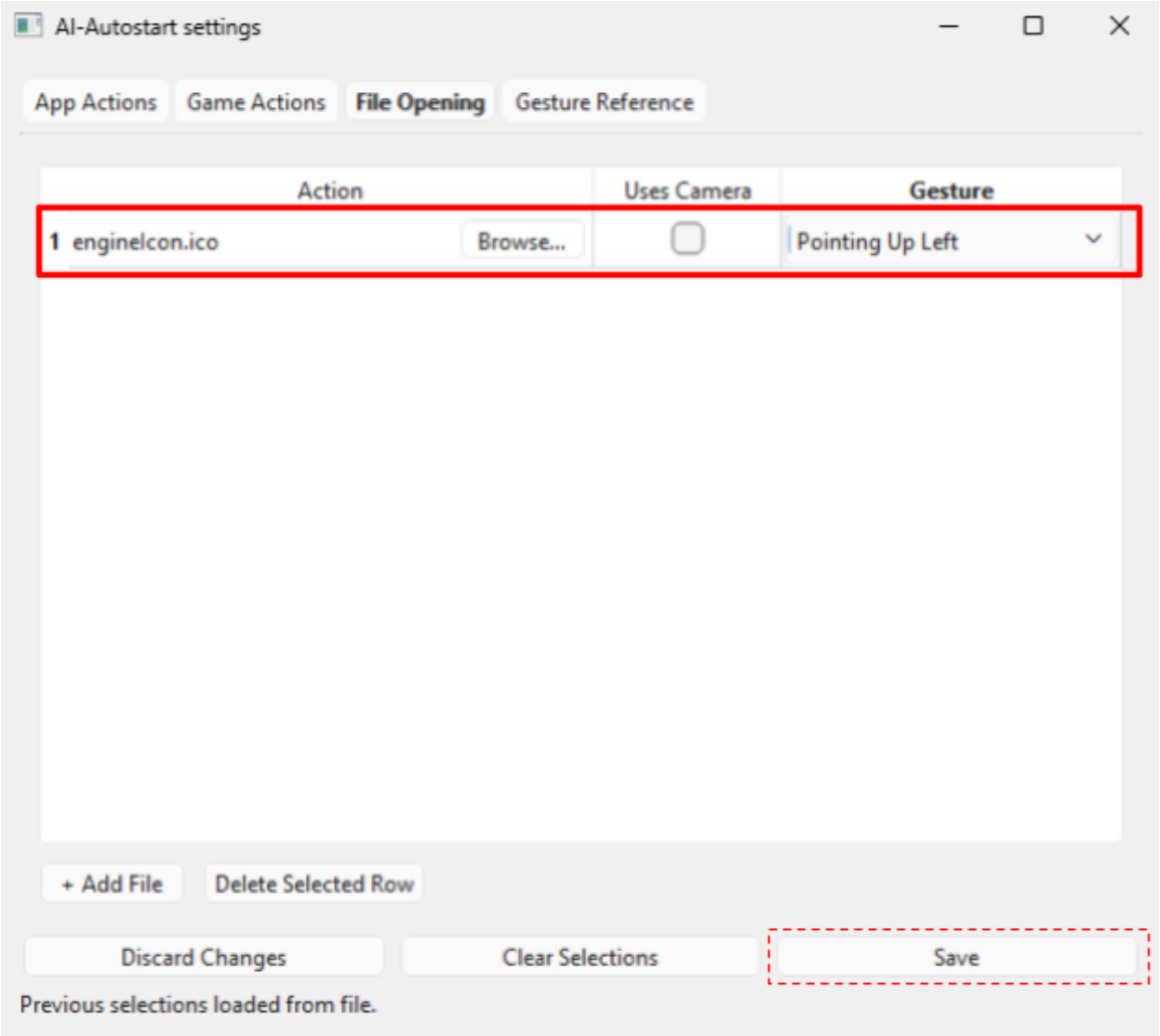
This feature allows users to open files or programs using gestures.

8.1 Adding a File

1. Open the settings window from the tray menu.
2. Go to **File Opening**.
3. Click + **Add File**.
4. A new row will appear.



5. Click **Browse** and select the file or program.
6. (Optional) Enable **Uses Camera** if the file will require the camera once opened.
7. Select a gesture.
8. Click **Save**.



The file will now open when the selected gesture is performed.

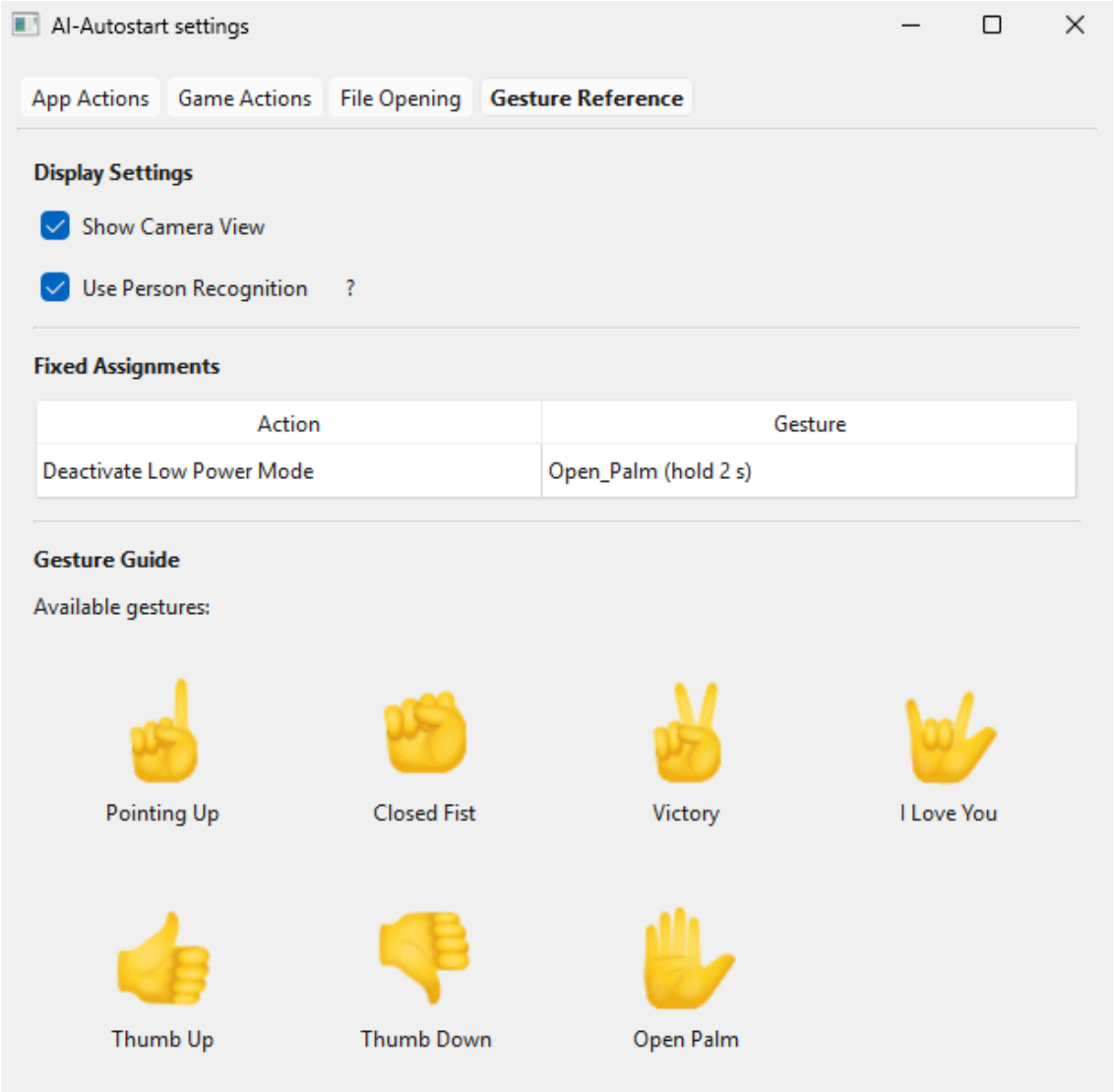
8.2 Deleting a File

1. Select the file row.
2. Click **Delete Selected Row**.
3. Click **Save**.

The file mapping will be removed.

9. Gesture Reference Page

The Gesture Reference page provides an overview of supported gestures and system behavior.



9.1 Display Settings

Show Camera View

- On: displays live camera preview
- Off: hides camera preview

Use Person Recognition

- On: focuses gesture detection on the interacting user (recommended in busy environments)
- Off: reduces computational load in private environments

9.2 Fixed Assignments

This section shows built-in gesture behaviors.

Example:

- **Open_Palm (hold 2 seconds)** → Deactivates low-power mode

These assignments cannot be changed.

9.3 Gesture Guide

The Gesture Guide displays supported gesture names and hand poses to help users perform gestures correctly.

10. Troubleshooting

Quick Triage

1. Restart AI-Autostart from the tray.
2. Ensure the webcam is not used by other applications.
3. Reopen Settings.
4. Reassign the gesture.
5. Click **Save**.
6. Test again.

Camera Preview Not Working

- Close other camera applications
- Reconnect the webcam
- Toggle **Show Camera View** off and on
- Restart gesture monitoring

Gestures Not Detected

- Improve lighting
- Keep the full hand visible
- Reduce background motion
- Enable **Use Person Recognition**

Actions Not Triggered

- Ensure gesture is not set to **None**
- Avoid duplicate gesture assignments
- Click **Save** after making changes

App Actions Issues

- Ensure application names match installed programs
- Reopen settings to refresh the app list

Game or File Launch Issues

- Re-select the file path
- Ensure the file still exists
- Use absolute paths

Low-Power Mode Confusion

- Perform **Open_Palm**
- Hold for **2 seconds**
- Wait for confirmation

This gesture is fixed and cannot be changed.

Settings Not Saving

- Click **Save**
- Check the confirmation message at the bottom of the window

Performance Issues

- Disable camera preview
- Disable person recognition in private environments
- Close unnecessary applications